Problem Set 2

Due on Canvas at noon Friday September 28th

Instructions: **No partial credit will be awarded.** All answers must be entered in the Problem Set 2 Quiz on Canvas. Note that it is not possible to edit your responses on Canvas after submitting, but you are able to submit a second time and your last submission is the one that will be graded. **Best practice is to complete the problem set on paper and then transfer solutions to Canvas.** You may collaborate on this problem set with other students in your learning team, but each student must submit his or her own solutions.

Grading: Grading is done on a ten point scale across all problems in the problem set. **Note that after submitting on canvas, you will see a score, but that does not include any points for any qualitative questions**. Grades will be finalized a few days after the due date.

Material covered: Lectures 6-8

1. “Cobb-Douglas” Production

A firm produced output using capital (K) and labor (L) according to the following production function:

$$Q = 3L^{1/3}K^{2/3}$$

where K and L are measured in hours (note: the “power rule” for derivatives applies even to fractional exponents). An hour of labor costs \$50 and an hour of capital costs \$25.

a. What is the optimal level of capital to employ as a function of labor?

b. Assume that the firm wishes to produce 300 units of output at minimum cost. Assuming it can freely vary its use of capital and labor, how much capital and labor should it employ? You may solve this problem using Excel and the Solver add-on. (If you feel like it, you can also solve this analytically, but this is not necessary for this part.)

c. What is the total cost of producing the 300 units?

d. What is the average cost of producing the 300 units?

e. Discuss your result, making sense of the capital-to-labor ratio.

f. What is the marginal product of labor relative to the price of labor evaluated at the quantity of labor you found in part a? (Use Solver.)

g. What is the marginal product of capital relative to the price of capital evaluated at the quantity of capital you found in part a? (Use Solver.)

h. Now suppose that the level of capital is fixed at the level you identified in part a, but you want to produce 350 units of output. How should you do this? What has happened to average cost? What has happened to the marginal products of capital and labor relative to their costs? (Again use Solver.) What is going on here?

2. Cost Functions

A young graduate with an eye for fashion has decided to start a business to make hand-knit wool sweaters. The weekly cost function to manufacture these sweaters is given by:

$$TC = 80 + q + 5q^2$$

a. What are your fixed costs?

b. What is the average fixed cost function? Explain what happens to average fixed costs as quantity increases?

c. What is the marginal cost function?

d. What is the average total cost function of the firm?

e. What is the average variable cost function?

f. Unfortunately, it appears as though the hand-knit sweater market looks a lot like "perfect competition". What would you expect the long run price to be for one of these sweaters?